

1 TLS

Sistema de 2 niveles

$\omega \left\{ \begin{array}{l} \text{---} |e\rangle \\ |g\rangle \text{---} \end{array} \right.$ Operadores de bajado y subida: $\sigma_- = |g\rangle\langle e|$, $\sigma_+ = |e\rangle\langle g|$
 $H = E_e |e\rangle\langle e| + E_g |g\rangle\langle g|$, $H \rightarrow H - E_g \Rightarrow H = \hbar\omega |e\rangle\langle e|$

Decaimiento: $K = \hbar\omega |e\rangle\langle e| - i\hbar\gamma |e\rangle\langle e|$

Evolución temporal de $|e\rangle$: $e^{-\frac{i}{\hbar} K t} |e\rangle = e^{-\frac{i}{\hbar} (\hbar\omega |e\rangle\langle e| - i\hbar\gamma |e\rangle\langle e|) t} |e\rangle = e^{-i\omega t} e^{-\gamma t} |e\rangle$

Ec. de Von Neumann: $\dot{\rho} = \frac{1}{i\hbar} [H, \rho] = \frac{1}{i\hbar} (H\rho - \rho H) = \frac{1}{i\hbar} (\hbar\omega |e\rangle\langle e|\rho - \hbar\omega \rho |e\rangle\langle e|)$

$\Rightarrow \dot{\rho} = \frac{1}{i\hbar} [K, \rho] = \frac{1}{i\hbar} (K\rho - \rho K^\dagger)$

$= \frac{1}{i\hbar} (\hbar\omega |e\rangle\langle e|\rho - i\hbar\gamma |e\rangle\langle e|\rho - \hbar\omega \rho |e\rangle\langle e| - i\hbar\gamma \rho |e\rangle\langle e|)$

$= \frac{1}{i\hbar} [H, \rho] - \gamma |e\rangle\langle e|\rho - \gamma \rho |e\rangle\langle e|$

$\downarrow$
 $\sigma_+ \sigma_- = |e\rangle\langle g|g\rangle\langle e| = |e\rangle\langle e|$

$= \frac{1}{i\hbar} [H, \rho] + \gamma (-\sigma_+ \sigma_- \rho - \rho \sigma_+ \sigma_-) \rightarrow \text{Falta un término!}$

$\text{Tr}(\dot{\rho}) \neq 0$, lo cual está mal pues la traza es invariante

$\Rightarrow \dot{\rho} = \mathcal{L}\rho = \frac{1}{i\hbar} [H, \rho] + \gamma (2\sigma_- \rho \sigma_+ - \sigma_+ \sigma_- \rho - \rho \sigma_+ \sigma_-)$

$\sigma_- \rho \sigma_+ = |g\rangle\langle e|\rho|e\rangle\langle g| = \langle e|\rho|e\rangle |g\rangle\langle g| \rightarrow \text{envía lo que esté en } |e\rangle \text{ a } |g\rangle$

Calculemos los eigenvalores de $\mathcal{L}$ para $|g\rangle\langle g|$, $|g\rangle\langle e|$, $|e\rangle\langle g|$ y $\sigma_z = |e\rangle\langle e| - |g\rangle\langle g|$:

$\mathcal{L}|g\rangle\langle g|$:

$\mathcal{L}|g\rangle\langle g| = \frac{1}{i\hbar} [H, |g\rangle\langle g|] + \gamma (2\sigma_- |g\rangle\langle g| \sigma_+ - \sigma_+ \sigma_- |g\rangle\langle g| - |g\rangle\langle g| \sigma_+ \sigma_-)$
 $= \frac{1}{i\hbar} (\hbar\omega |e\rangle\langle e|g\rangle\langle g| - \hbar\omega |g\rangle\langle g|e\rangle\langle e|)$
 $+ \gamma (2|g\rangle\langle e|g\rangle\langle g|e\rangle\langle g| - |e\rangle\langle g|g\rangle\langle e|g\rangle\langle g| - |g\rangle\langle g|e\rangle\langle g|g\rangle\langle e|)$
 $= 0 \Rightarrow \lambda_{gg} = 0$

$\mathcal{L}|g\rangle\langle e|$:

$\mathcal{L}|g\rangle\langle e| = \frac{1}{i\hbar} [H, |g\rangle\langle e|] + \gamma (2\sigma_- |g\rangle\langle e| \sigma_+ - \sigma_+ \sigma_- |g\rangle\langle e| - |g\rangle\langle e| \sigma_+ \sigma_-)$

$$\begin{aligned}
&= \frac{1}{i\hbar} (\hbar\omega |e\rangle\langle e|g\rangle\langle e| - \hbar\omega |g\rangle\langle e|e\rangle\langle e|) \\
&\quad + \tau (2|g\rangle\langle e|g\rangle\langle e|e\rangle\langle g| - |e\rangle\langle g|g\rangle\langle e|g\rangle\langle e| - |g\rangle\langle e|e\rangle\langle g|g\rangle\langle e|) \\
&= i\omega |g\rangle\langle e| - \tau |g\rangle\langle e| = (i\omega - \tau) |g\rangle\langle e| \Rightarrow \lambda_{ge} = i\omega - \tau
\end{aligned}$$

$\mathcal{L}|e\rangle\langle g|$:

$$\begin{aligned}
\mathcal{L}|e\rangle\langle g| &= \frac{1}{i\hbar} [H, |e\rangle\langle g|] + \tau (2\sigma_- |e\rangle\langle g|\sigma_+ - \sigma_+ \sigma_- |e\rangle\langle g| - |e\rangle\langle g|\sigma_+ \sigma_-) \\
&= \frac{1}{i\hbar} (\hbar\omega |e\rangle\langle e|e\rangle\langle g| - \hbar\omega |e\rangle\langle g|e\rangle\langle e|) \\
&\quad + \tau (2|g\rangle\langle e|e\rangle\langle g|e\rangle\langle g| - |e\rangle\langle g|g\rangle\langle e|e\rangle\langle g| - |e\rangle\langle g|e\rangle\langle g|g\rangle\langle e|) \\
&= -i\omega |e\rangle\langle g| - \tau |e\rangle\langle g| = -(i\omega + \tau) |e\rangle\langle g| \Rightarrow \lambda_{eg} = -(i\omega + \tau)
\end{aligned}$$

$\mathcal{L}\sigma_z$:

$$\begin{aligned}
\mathcal{L}\sigma_z &= \frac{\omega}{i} (|e\rangle\langle e|e\rangle\langle e| - |e\rangle\langle e|g\rangle\langle g| - |e\rangle\langle e|e\rangle\langle e| + |g\rangle\langle g|e\rangle\langle e|) \\
&\quad + \tau (2|g\rangle\langle e|e\rangle\langle e|e\rangle\langle g| - 2|g\rangle\langle e|g\rangle\langle g|e\rangle\langle g| - |e\rangle\langle g|g\rangle\langle e|e\rangle\langle e| \\
&\quad + |e\rangle\langle g|g\rangle\langle e|g\rangle\langle g| - |e\rangle\langle e|e\rangle\langle g|g\rangle\langle e| + |g\rangle\langle g|e\rangle\langle g|g\rangle\langle e|) \\
&= \tau (2|g\rangle\langle g| - |e\rangle\langle e| - |e\rangle\langle e|) = -2\tau (|e\rangle\langle e| - |g\rangle\langle g|) \Rightarrow \lambda_{\sigma_z} = -2\tau
\end{aligned}$$

Eigensistema izquierdo:

Ahora queremos resolver el sist. de eigenvectores izquierdos $\mathcal{L}^+ \check{f} = \lambda^* \check{f}$.

¿Qué es $\mathcal{L}^+$?

Supongamos un operador $\Theta f = A f B$

$$(\mathcal{f}_1, \Theta \mathcal{f}_2) = (\Theta^\dagger \mathcal{f}_1, \mathcal{f}_2)$$

↳ producto interno de Hilbert-Schmidt $(A, B)_{HS} = \text{Tr}(A^\dagger B)$

$$\Rightarrow \text{Tr}(\mathcal{f}_1^\dagger \Theta \mathcal{f}_2) = \text{Tr}((\Theta^\dagger \mathcal{f}_1)^\dagger \mathcal{f}_2) \Rightarrow \text{Tr}(\mathcal{f}_1^\dagger A \mathcal{f}_2 B) = \text{Tr}(B \mathcal{f}_1^\dagger A \mathcal{f}_2) = \text{Tr}((A^\dagger \mathcal{f}_1 B^\dagger)^\dagger \mathcal{f}_2)$$

$$\Rightarrow \Theta^\dagger \mathcal{f}_1 = A^\dagger \mathcal{f}_1 B^\dagger$$

$$\Rightarrow \mathcal{L}^+ f = \frac{i}{\hbar} [H, f] + \sum_i \frac{\Gamma_i}{2} (2A_i^\dagger f A_i - A_i^\dagger A f - f A_i^\dagger A_i)$$

$\mathcal{L}^+ |g\rangle\langle g|$:

$$\sigma_+^\dagger = \sigma_- , \sigma_-^\dagger = \sigma_+$$

$$\begin{aligned}
\mathcal{L}^+ |g\rangle\langle g| &= \frac{i}{\hbar} [\hbar\omega |e\rangle\langle e|, |g\rangle\langle g|] + \tau (2\sigma_+ |g\rangle\langle g|\sigma_- - \sigma_+ \sigma_- |g\rangle\langle g| - |g\rangle\langle g|\sigma_+ \sigma_-) \\
&= \frac{i}{\hbar} (\hbar\omega |e\rangle\langle e|g\rangle\langle g| - \hbar\omega |g\rangle\langle g|e\rangle\langle e|) + 2\tau |e\rangle\langle g|g\rangle\langle g|g\rangle\langle e| \\
&\quad - \tau |e\rangle\langle g|g\rangle\langle e|g\rangle\langle g| - \tau |g\rangle\langle g|e\rangle\langle g|g\rangle\langle e|
\end{aligned}$$

$$= 2\tau |exel| \therefore |g \times g| \text{ no es eigenvector}$$

$$\hat{I}^+ |g \times e|:$$

$$\begin{aligned} \hat{I}^+ |g \times e| &= \frac{i}{\hbar} (\hbar \omega |exel|g \times e| - \hbar \omega |g \times e|exel) + 2\tau |exg|g \times e|g \times e| \\ &\quad - \tau |exg|g \times e|g \times e| - \tau |g \times e|exg|g \times e| \\ &= -i\omega |g \times e| - \tau |g \times e| \Rightarrow \lambda_{ge}^* = -(i\omega + \tau) \end{aligned}$$

$$\hat{I}^+ |exg|:$$

$$\begin{aligned} \hat{I}^+ |exg| &= \frac{i}{\hbar} (\hbar \omega |exel|exg| - \hbar \omega |exg|exel) + 2\tau |exg|exg|g \times e| \\ &\quad - \tau |exg|g \times e|exg| - \tau |exg|exg|g \times e| \\ &= i\omega |exg| - \tau |exg| \Rightarrow \lambda_{eg}^* = i\omega - \tau \end{aligned}$$

$$\hat{I}^+ |exel|:$$

$$\begin{aligned} \hat{I}^+ |exel| &= \frac{i}{\hbar} (\hbar \omega |exel|exel - \hbar \omega |exel|exel) + 2\tau |exg|exelg \times e| \\ &\quad - \tau |exg|g \times e|exel - \tau |exel|exg|g \times e| = -2\tau |exel| \Rightarrow \lambda_{ee}^* = -2\tau \end{aligned}$$

$$\hat{I}^+ \mathbb{1}:$$

$$|g\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |e\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow \mathbb{1} = |exel| + |g \times g|$$

$$\begin{aligned} \hat{I}^+ \mathbb{1} &= 2\tau |exg|exelg \times e| + 2\tau |exg|g \times g|g \times e| - \tau |exg|g \times e|exel - \tau |exg|g \times e|g \times g| \\ &\quad - \tau |exel|exg|g \times e| - \tau |g \times g|exg|g \times e| \\ &= 2\tau |exel| - \tau |exel| - \tau |exel| = 0 \Rightarrow \lambda_{\mathbb{1}}^* = 0 \end{aligned}$$

Tabla de eigenvalores izquierdos y derechos:

λ	λ^*	$\hat{F}$	$\check{F}$	$\sigma_- \sigma_+ = g \times g = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$
0	0	$ g \times g $	$\mathbb{1}$	$\sigma_+ \sigma_- = exel = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$
-2τ	-2τ	σ_z	$ exel $	$\sigma_- = g \times e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$
$-\tau + i\omega$	$-\tau - i\omega$	$ g \times e $	$ g \times e $	$\sigma_+ = exg = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$
$-\tau - i\omega$	$-\tau + i\omega$	$ exg $	$ exg $	

$$\begin{aligned} (\hat{F}, \check{F})_{\text{HS}} &= \text{Tr}(\hat{F}^+ \check{F}) = 1, \quad \text{ej: } \text{Tr}(|g \times e|^\top |g \times e|) = \text{Tr}\left(\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right) \\ &= \text{Tr}\left(\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right) = 1 \quad \checkmark \end{aligned}$$

Vectorización

$$F = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow F_v = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}, \quad A \otimes B = (A \otimes B^T) F_v$$

$$|g \times g\rangle_v = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |e \times e\rangle_v = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad |g \times e\rangle_v = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |e \times g\rangle_v = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$H = \hbar \omega |e \times e\rangle$$

$$\sigma_- \sigma_+ = |g \times g\rangle = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\sigma_+ \sigma_- = |e \times e\rangle = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\sigma_- = |g \times e\rangle = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\sigma_+ = |e \times g\rangle = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\mathcal{L} F = \frac{1}{i\hbar} [H, F] + \gamma (2\sigma_- F \sigma_+ - \sigma_+ \sigma_- F - F \sigma_+ \sigma_-)$$

$$= -i\omega (\sigma_+ \sigma_- F - F \sigma_+ \sigma_-) + \gamma (2\sigma_- F \sigma_+ - \sigma_+ \sigma_- F - F \sigma_+ \sigma_-)$$

$$\Rightarrow \mathcal{L} F_v = -i\omega [(\overbrace{\sigma_+ \sigma_-}^{\text{hermiticos}} \otimes \mathbb{1}) F_v - (\mathbb{1} \otimes \sigma_+ \sigma_-) F_v] + \gamma [2(\sigma_- \otimes \sigma_+^T) F_v - (\sigma_+ \sigma_- \otimes \mathbb{1}) F_v - (\mathbb{1} \otimes \sigma_+ \sigma_-) F_v]$$

$$= -i\omega \left[\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} F_v - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} F_v \right] + \gamma \left[2 \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} F_v - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} F_v - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} F_v \right]$$

Calculemos los eigenvalores derechos:

$$\mathcal{L} |g \times g\rangle = -i\omega \left[\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right] + \gamma \left[2 \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right] = 0 \Rightarrow \lambda_{gg} = 0$$

$$\mathcal{L} |g \times e\rangle = -i\omega \left[\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right] + \gamma \left[2 \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right]$$

$$= i\omega \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} - \gamma \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \lambda_{ge} = i\omega - \gamma$$

$$\mathcal{L}|e \times g\rangle = -i\omega \left[\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right] + \gamma \left[2 \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right]$$

$$= -i\omega \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} - \gamma \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \Rightarrow \lambda_{eg} = -i\omega - \gamma$$

$$\mathcal{L}\sigma_z = -i\omega \left[\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right] + \gamma \left[2 \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right] = \gamma \left[2 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} - 2 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right]$$

$$= -2\gamma \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \Rightarrow \lambda_{\sigma_z} = -2\gamma$$

Todos los eigenvalores concuerdan con los obtenidos anteriormente

$$\mathcal{L}(\rho) = -\frac{i}{\hbar} [H, \rho] + \gamma (2\sigma\rho\sigma^\dagger - \rho\sigma^\dagger\sigma - \sigma^\dagger\sigma\rho)$$

$$K = H - i\hbar\gamma\sigma^\dagger\sigma$$

$$\begin{aligned} \Rightarrow -\frac{i}{\hbar} [K, \rho] &= -\frac{i}{\hbar} (K\rho - \rho K^\dagger) = -\frac{i}{\hbar} \{ (H - i\hbar\gamma\sigma^\dagger\sigma)\rho - \rho(H + i\hbar\gamma\sigma^\dagger\sigma) \} \\ &= -\frac{i}{\hbar} (H\rho - i\hbar\gamma\sigma^\dagger\sigma\rho - \rho H - i\hbar\gamma\rho\sigma^\dagger\sigma) \\ &= -\frac{i}{\hbar} [H, \rho] - \gamma\sigma^\dagger\sigma\rho - \gamma\rho\sigma^\dagger\sigma \end{aligned}$$

$$\Rightarrow \mathcal{L}\rho = -\frac{i}{\hbar} [K, \rho] + 2\gamma\sigma\rho\sigma^\dagger = -\frac{i}{\hbar} (K\rho - \rho K^\dagger) + 2\gamma\sigma\rho\sigma^\dagger$$

$$\mathcal{L} = -\frac{i}{\hbar} (K \otimes \mathbb{1} - \mathbb{1} \otimes K^\dagger) + 2\gamma\sigma \otimes \sigma$$

$$= (\sigma \otimes \sigma^\dagger) \rho_\nu = (\sigma \otimes \sigma) \rho_\nu$$

Desde 0:

$$H = \hbar\omega\sigma^\dagger\sigma, \quad K = \hbar\omega\sigma^\dagger\sigma - i\hbar\gamma\sigma^\dagger\sigma$$

$$\begin{aligned} \sigma^\dagger &= \sigma_+ \\ \sigma &= \sigma_- \end{aligned}$$

hamiltoniano efectivo

$$K = \hbar\omega \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} - i\hbar\gamma \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \hbar \begin{pmatrix} 0 & 0 \\ 0 & \omega - i\gamma \end{pmatrix} \Rightarrow K^\dagger = \hbar \begin{pmatrix} 0 & 0 \\ 0 & \omega + i\gamma \end{pmatrix}$$

$$K \otimes 1 = \hbar \begin{pmatrix} 0 & 0 \\ 0 & \omega - i\gamma \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \hbar \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \omega - i\gamma & 0 \\ 0 & 0 & 0 & \omega - i\gamma \end{pmatrix}$$

$$1 \otimes K^\dagger = \hbar \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 0 \\ 0 & \omega + i\gamma \end{pmatrix} = \hbar \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \omega + i\gamma & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \omega + i\gamma \end{pmatrix}$$

$$2\gamma \sigma \otimes \sigma = 2\gamma \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = 2\gamma \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow \mathcal{L} = -i \left\{ \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \omega - i\gamma & 0 \\ 0 & 0 & 0 & \omega - i\gamma \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \omega + i\gamma & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \omega + i\gamma \end{pmatrix} \right\} + \begin{pmatrix} 0 & 0 & 0 & 2\gamma \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow \mathcal{L} = \begin{pmatrix} 0 & 0 & 0 & 2\gamma \\ 0 & -\gamma + i\omega & 0 & 0 \\ 0 & 0 & -\gamma - i\omega & 0 \\ 0 & 0 & 0 & -2\gamma \end{pmatrix}$$

Vemos que $\mathcal{L}$ es una Matriz triangular superior así que los eigenvalores están en la diagonal.

$$\Rightarrow \lambda_1 = 0, \lambda_2 = -2\gamma, \lambda_3 = -\gamma + i\omega, \lambda_4 = -\gamma - i\omega \rightarrow \text{eigenvalores de } \mathcal{L}$$

$$\lambda_1 = 0: \quad \mathcal{L} \mathbf{g} = 0 \mathbf{g} \Rightarrow \begin{pmatrix} 0 & 0 & 0 & 2\gamma \\ 0 & -\gamma + i\omega & 0 & 0 \\ 0 & 0 & -\gamma - i\omega & 0 \\ 0 & 0 & 0 & -2\gamma \end{pmatrix} \begin{pmatrix} g_{00} \\ g_{01} \\ g_{10} \\ g_{11} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} 2\gamma g_{11} = 0 \\ (-\gamma + i\omega) g_{01} = 0 \\ (-\gamma - i\omega) g_{10} = 0 \\ -2\gamma g_{11} = 0 \end{cases} \Rightarrow \begin{cases} g_{00} = ? \\ g_{01} = 0 \\ g_{10} = 0 \\ g_{11} = 0 \end{cases}$$

$$\text{Sea } g_{00} = 1$$

$$\Rightarrow \hat{g}_{\lambda_1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \sigma \sigma^\dagger = |g \times g\rangle$$

$$\lambda_2 = -2\gamma$$

$$\mathcal{L} \mathbf{g} = -2\gamma \mathbf{g} \Rightarrow \begin{pmatrix} 0 & 0 & 0 & 2\gamma \\ 0 & -\gamma + i\omega & 0 & 0 \\ 0 & 0 & -\gamma - i\omega & 0 \\ 0 & 0 & 0 & -2\gamma \end{pmatrix} \begin{pmatrix} g_{00} \\ g_{01} \\ g_{10} \\ g_{11} \end{pmatrix} = -2\gamma \begin{pmatrix} g_{00} \\ g_{01} \\ g_{10} \\ g_{11} \end{pmatrix}$$

$$\Rightarrow \begin{cases} 2\gamma g_{11} = -2\gamma g_{00} \Rightarrow g_{00} = -g_{11} \\ (-\gamma + i\omega) g_{01} = -2\gamma g_{01} \Rightarrow g_{01} = 0 \\ (-\gamma - i\omega) g_{10} = -2\gamma g_{10} \Rightarrow g_{10} = 0 \\ -2\gamma g_{11} = -2\gamma g_{11} \Rightarrow g_{11} = ? \end{cases}$$

See $g_{11} = 1 \Rightarrow g_{00} = -1$

$$\Rightarrow \hat{g}_{\lambda_2} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \sigma_z = |e\rangle\langle e| - |g\rangle\langle g| = \sigma^+ \sigma - \sigma \sigma^+$$

$\lambda_3 = -\gamma + i\omega$:

$$\mathcal{L}g = (-\gamma + i\omega)g \Rightarrow \begin{pmatrix} 0 & 0 & 0 & 2\gamma \\ 0 & -\gamma + i\omega & 0 & 0 \\ 0 & 0 & -\gamma - i\omega & 0 \\ 0 & 0 & 0 & -2\gamma \end{pmatrix} \begin{pmatrix} g_{00} \\ g_{01} \\ g_{10} \\ g_{11} \end{pmatrix} = (-\gamma + i\omega) \begin{pmatrix} g_{00} \\ g_{01} \\ g_{10} \\ g_{11} \end{pmatrix}$$

$$\Rightarrow \begin{cases} 2\gamma g_{11} = (-\gamma + i\omega)g_{00} \Rightarrow g_{00} = 0 \\ (-\gamma + i\omega)g_{01} = (-\gamma + i\omega)g_{01} \Rightarrow g_{01} = ? \\ (-\gamma - i\omega)g_{10} = (-\gamma + i\omega)g_{10} \Rightarrow g_{10} = 0 \\ -2\gamma g_{11} = (-\gamma + i\omega)g_{11} \Rightarrow g_{11} = 0 \end{cases}$$

See $g_{01} = 1$

$$\Rightarrow \hat{g}_{\lambda_3} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \sigma = |g\rangle\langle e|$$

$\lambda_4 = -\gamma - i\omega$:

$$\mathcal{L}g = (-\gamma - i\omega)g \Rightarrow \begin{pmatrix} 0 & 0 & 0 & 2\gamma \\ 0 & -\gamma + i\omega & 0 & 0 \\ 0 & 0 & -\gamma - i\omega & 0 \\ 0 & 0 & 0 & -2\gamma \end{pmatrix} \begin{pmatrix} g_{00} \\ g_{01} \\ g_{10} \\ g_{11} \end{pmatrix} = (-\gamma - i\omega) \begin{pmatrix} g_{00} \\ g_{01} \\ g_{10} \\ g_{11} \end{pmatrix}$$

$$\Rightarrow \begin{cases} 2\gamma g_{11} = (-\gamma - i\omega)g_{00} \Rightarrow g_{00} = 0 \\ (-\gamma + i\omega)g_{01} = (-\gamma - i\omega)g_{01} \Rightarrow g_{01} = 0 \\ (-\gamma - i\omega)g_{10} = (-\gamma - i\omega)g_{10} \Rightarrow g_{10} = ? \\ -2\gamma g_{11} = (-\gamma - i\omega)g_{11} \Rightarrow g_{11} = 0 \end{cases}$$

See $g_{10} = 1 \Rightarrow \hat{g}_{\lambda_4} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \sigma^+ = |e\rangle\langle g|$

$$H = E_e |e\rangle\langle e| + E_g |g\rangle\langle g|, \quad \hbar\omega = E_e - E_g \Rightarrow E_e = \hbar\omega + E_g$$

$$= \hbar\omega |e\rangle\langle e| + E_g |e\rangle\langle e| + E_g |g\rangle\langle g|$$

$$= \hbar\omega |e\rangle\langle e| + E_g (|e\rangle\langle e| + |g\rangle\langle g|)$$

$$= \hbar\omega |e\rangle\langle e| + E_g \mathbb{1}$$

Solución temporal:

$$\dot{\rho}(t) = \mathcal{L} \rho(t) \Rightarrow \rho(t) = e^{\mathcal{L}t} \rho(0) \quad \text{independiente de } t$$

$$\text{Expandiendo } \rho(t): \rho(t) = \sum_n c_n \hat{\rho}_n \Rightarrow \dot{\rho}(t) = \sum_n \dot{c}_n \hat{\rho}_n$$

$$\Rightarrow \sum_n \dot{c}_n \hat{\rho}_n = \mathcal{L} \sum_n c_n \hat{\rho}_n = \sum_n c_n \mathcal{L} \hat{\rho}_n = \sum_n c_n \lambda_n \hat{\rho}_n$$

$$\Rightarrow \sum_n \dot{c}_n = \sum_n c_n \lambda_n \Rightarrow \dot{c}_n = \lambda_n c_n \Rightarrow c_n(t) = e^{\lambda_n t} c_n(0)$$

$$\Rightarrow \rho(t) = \sum_n c_n(0) e^{\lambda_n t} \hat{\rho}_n$$

$$\hookrightarrow c_n(0) = \text{Tr}(\hat{\rho}_n^\dagger \rho(0))$$

$$\text{Si } \rho(0) = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix}$$

$$\Rightarrow c_1(0) = \rho_{00} + \rho_{11}$$

$$c_2(0) = \rho_{01}$$

$$c_3(0) = \rho_{10}$$

$$c_4(0) = \rho_{11}$$

$$\begin{aligned} \Rightarrow \rho(t) &= (\underbrace{\rho_{00} + \rho_{11}}_1) e^{\underbrace{\mathcal{L}}_0 t} |g\rangle\langle g| + \rho_{01} e^{(-\gamma+i\omega)t} |g\rangle\langle e| + \rho_{10} e^{(-\gamma-i\omega)t} |e\rangle\langle g| + \rho_{11} e^{-2\gamma t} \sigma_z \\ &= |g\rangle\langle g| + \rho_{01} e^{(-\gamma+i\omega)t} |g\rangle\langle e| + \rho_{10} e^{(-\gamma-i\omega)t} |e\rangle\langle g| + \rho_{11} e^{-2\gamma t} |e\rangle\langle e| - \rho_{11} e^{-2\gamma t} |g\rangle\langle g| \\ &= (1 - \rho_{11} e^{-2\gamma t}) |g\rangle\langle g| + \rho_{01} e^{(-\gamma+i\omega)t} |g\rangle\langle e| + \rho_{10} e^{(-\gamma-i\omega)t} |e\rangle\langle g| + \rho_{11} e^{-2\gamma t} |e\rangle\langle e| \\ &= \begin{pmatrix} 1 - \rho_{11} e^{-2\gamma t} & \rho_{01} e^{(-\gamma+i\omega)t} \\ \rho_{10} e^{(-\gamma-i\omega)t} & \rho_{11} e^{-2\gamma t} \end{pmatrix} \end{aligned}$$